# DIGIPIN-Based Intelligent Merchant Identification for QR-less UPI Payments with Context-Aware Expense Analysis

*Enabling QR-less UPI Payments through Location Intelligence and Merchant Prediction*


Name: **Supreeth Kumar Jagarlamudi**

Register No: **192372021**

Department: **Computer Science and Engineering (AI)**

Guide Name: **Dr. Anoop P**

Institution: **Saveetha School of Engineering**


***Product Design and Developement***

*March 2026*

# Abstract

This project presents a system for enabling QR-less UPI payments through intelligent merchant identification using location intelligence and contextual data. Current digital payment methods rely heavily on manual QR code scanning, which reduces user experience and introduces friction in environments with high merchant density or limited user interaction time.

The proposed approach leverages mobile device location inputs along with contextual signals to estimate the user's position and map it to a spatial representation such as DIGIPIN. Based on this mapping, a set of candidate merchants is identified within the user’s proximity. A decision layer is then applied to evaluate and rank these merchants using relevant contextual features, enabling the system to predict the most probable merchant for a transaction.

The system utilizes UPI deep linking to initiate payment by redirecting users to existing UPI applications such as PhonePe or Google Pay. This approach maintains compatibility with the existing UPI ecosystem while introducing an intelligent merchant identification layer prior to payment initiation.

In addition to merchant identification, the system extends its functionality by incorporating a context-aware expense intelligence module. By leveraging inferred merchant information along with spatial and temporal context, the system enables automatic classification of transactions and provides basic insights into user spending patterns. This component demonstrates how pre-payment intelligence can be further utilized to enhance post-transaction financial awareness.

The work further examines key challenges such as location uncertainty, multi-merchant ambiguity within dense spatial regions, and limitations associated with payment initiation mechanisms, forming the basis for system design considerations and future enhancements.

# Problem Statement

Digital payments in India have seen widespread adoption with the growth of UPI-based applications such as PhonePe and Google Pay.A majority of in-person transactions rely on QR code scanning, where users manually open a payment application and scan a merchant's QR code to initiate a transaction. While effective, this process introduces friction, especially in environments with high merchant density, limited space, or time-sensitive interactions.

In scenarios such as small retail clusters, food courts, or multi-store complexes, multiple merchants may exist within close physical proximity. Users are required to identify the correct QR code among several options, which can lead to confusion, delays, or incorrect payments. Additionally, QR-based interactions depend on clear visibility and accessibility of the code, which may not always be feasible.

With the emergence of spatial addressing systems such as DIGIPIN, there exists an opportunity to explore location-driven approaches for identifying merchants. However, challenges arise due to limitations in location accuracy, especially in indoor or dense urban environments, and the inability of spatial grids to distinguish between multiple merchants within the same region or across different floors.

Furthermore, current payment workflows lack contextual intelligence and rely entirely on manual user input for merchant selection. There is no mechanism to infer user intent based on location, movement, or historical interaction patterns.

In addition, existing digital payment applications provide limited support for understanding user spending behavior beyond basic transaction history. Users are required to manually interpret their expenses, and there is no automated mechanism to classify transactions or derive insights based on contextual information such as merchant type, location, or time of transaction.

These limitations highlight the need for a system that can intelligently identify nearby merchants and reduce dependency on manual QR code scanning, while remaining compatible with existing UPI infrastructure.

# System Overview

The system is designed as a location-driven pipeline that transforms raw user context into a payment-ready action without requiring manual QR code interaction.

At a high level, the workflow can be represented as a sequence of stages:

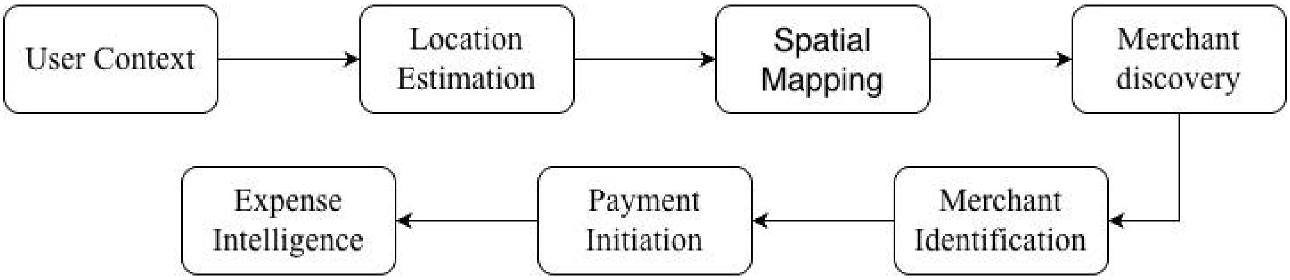


Fig.1. Workflow Diagram

The process begins when the user opens the application, at which point the system captures the user's current context, including location and other sensor available environmental signals. This information is used to estimate the user's position in real time. The estimated location is then mapped to a spatial representation(DIGIPIN - An open source grid addressing system), which acts as a reference for identifying nearby merchants. Based on this mapped region, a set of candidate merchants is retrieved from the underlying merchant data layer.

Since multiple merchants may exist within the same spatial region, the system introduces a merchant identification stage. In this stage, the candidate merchants are evaluated based on contextual relevance, and the system determines the most probable merchant associated with the user's intent. Once a target merchant is identified, the system redirects the user to an external UPI application for transaction initiation.

In addition to the primary payment flow, the system incorporates a post-transaction processing stage in which the inferred merchant information and contextual data are utilized to enable automatic classification of expenses. This stage allows the system to derive basic insights into user spending behavior without requiring manual input, thereby extending the utility of the merchant identification process beyond payment initiation.

This pipeline shifts the interaction model from manual QR scanning to an intelligent, context-aware identification process, forming the core foundation of the proposed system.

# Merchant Identification Flow

One of the central challenges in the proposed system is identifying the correct merchant within a given spatial region. In many real-world environments, multiple merchants may exist within close proximity or even within the same spatial grid, making direct location-based selection insufficient.

The merchant identification process is designed to resolve ambiguity in environments where multiple merchants may exist within close physical proximity. While spatial mapping techniques such as DIGIPIN help reduce the search area to a fine-grained region(approximately 4m x 4m), this alone is not sufficient to uniquely identify a single merchant.

The process begins by mapping the user's estimated location to a corresponding DIGIPIN cell. This step significantly narrows down the search space by restricting the system to a small spatial region. Based on this mapping, a set of candidate merchants associated with the identified DIGIPIN cell is retrieved.

In scenarios where multiple merchants exist within the same cell, the system proceeds to an ambiguity resolution stage. At this stage, the candidate merchants are further evaluated using additional contextual and spatial indicators. These may include relative proximity within the cell, user orientation, temporal context, and prior interaction patterns.

Following this evaluation, each candidate merchant is assigned a relative likelihood score representing its relevance to the user's current intent. The system then selects the highest-ranked merchant as the predicted target, or alternatively presents a small set of top-ranked merchants to the user for confirmation.

By combining fine-grained spatial mapping with contextual evaluation, the system aims to reduce ambiguity while maintaining flexibility in handling real-world complexities such as dense merchant environments and multi-level structures

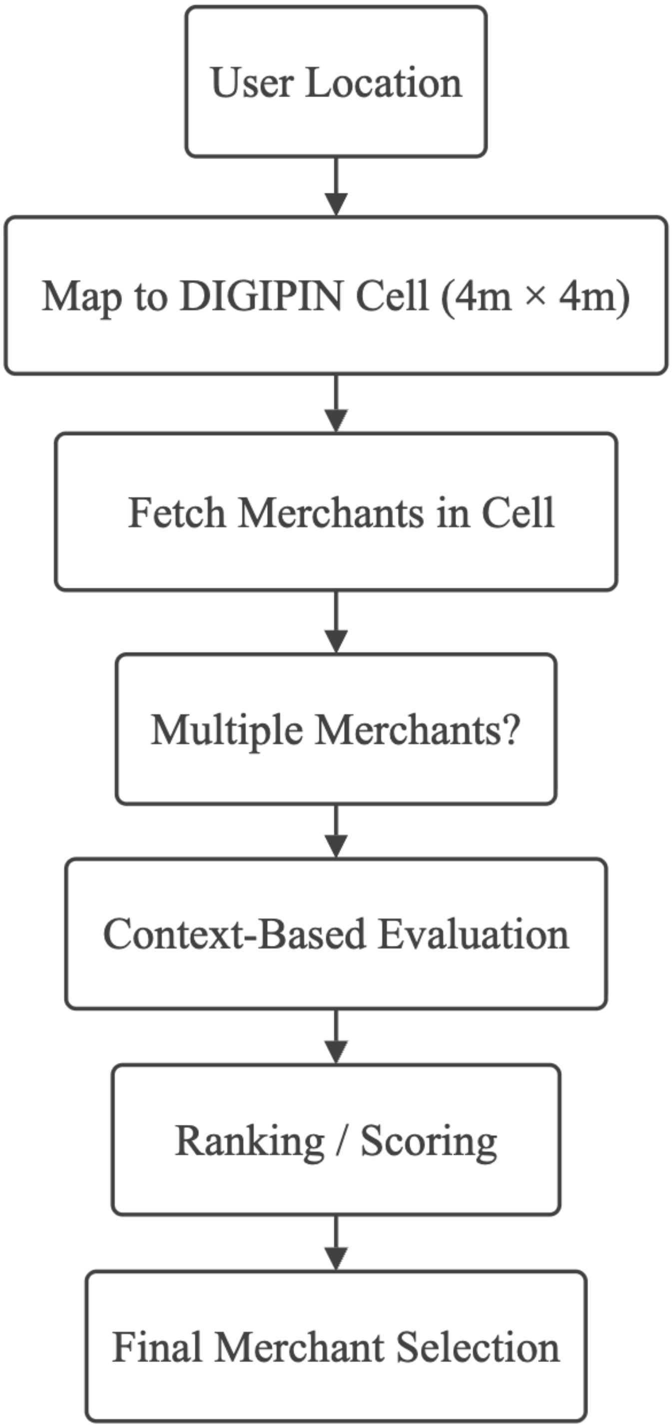


Fig.2. DIGIPIN-Based Merchant Identification Flow

# Proposed Implementation Approach

This section outlines the practical approach for implementing the proposed QR-less merchant identification system along with the context-aware expense intelligence module. The implementation is designed to be lightweight, modular, and compatible with existing mobile and UPI ecosystems.

## 1. System Architecture

The system is structured into multiple functional components that interact in a sequential pipeline:

- Mobile Application (iOS): Acts as the primary user interface, responsible for capturing user context such as location and initiating the workflow.
- Location Service Module: Retrieves real-time user location using device GPS and available sensor data.
- DIGIPIN Mapping Module: Converts geographic coordinates into a corresponding DIGIPIN cell to enable fine-grained spatial identification.
- Merchant Data Layer: Stores merchant information associated with DIGIPIN cells. This can be implemented using a local database or a cloud-based datastore.
- Ranking Engine: Evaluates candidate merchants using contextual features and assigns a relevance score to each merchant.
- UPI Integration Module: Generates and triggers UPI deep links to initiate payment through external applications.
- Expense Intelligence Module: Processes transaction context to classify expenses and derive basic insights.

## 2. Technology Stack

The implementation leverages commonly available technologies to ensure feasibility and maintainability:

- Frontend: Swift (iOS development using UIKit or SwiftUI)
- Backend: Python (using frameworks such as Flask or FastAPI for handling application logic and APIs)
- Database: MySQL for structured storage of merchant data and related attributes
- APIs: Location services (Core Location), mapping services, and UPI deep linking

The use of a relational database such as MySQL is suitable for managing structured merchant information, including location coordinates, category, and identifiers. It enables efficient querying and ensures data consistency, making it appropriate for the prototype implementation despite the availability of NoSQL alternatives

## 3. Merchant Data Management

Merchant data is managed through a structured onboarding process, where merchants register within the system by providing essential details such as business name, category, and UPI identifier. During registration, the merchant's location is captured using device-based location services, and the corresponding DIGIPIN is generated and stored.

Each merchant entry includes attributes such as geographic coordinates, DIGIPIN mapping, category, and payment identifier. This structured data is maintained within a relational database (MySQL), enabling efficient retrieval of merchants associated with a specific spatial region.

This approach ensures that the merchant dataset remains scalable and self-maintained, reducing dependency on external data sources. It also allows the system to dynamically update and expand its coverage as new merchants are onboarded. In a full-scale deployment, additional verification mechanisms and integration with external merchant registries can be incorporated to ensure data accuracy and reliability

## 4. Ranking Logic Execution

The ranking engine operates using a lightweight scoring mechanism that evaluates contextual parameters such as proximity, orientation, interaction history, and temporal relevance. This computation can be performed either locally on the device or through a backend service, depending on system design requirements.

The modular design allows flexibility in refining the ranking logic without affecting other components of the system.

## 5. Expense Intelligence Implementation

The expense intelligence module utilizes the identified merchant and associated contextual data to automatically classify transactions into predefined categories such as food, transport, or retail. This classification is primarily rule-based and can be enhanced using simple machine learning techniques in future iterations.

The module further provides basic insights into user spending patterns, demonstrating the extended utility of the merchant identification process

## 6. Implementation Considerations

The proposed implementation prioritizes simplicity and feasibility while maintaining extensibility. Key considerations include handling location inaccuracies, ensuring low-latency decision making, and maintaining compatibility with existing UPI applications.

The modular architecture ensures that each component can be independently improved or replaced as the system evolves.

# Key Challenges and Limitations

While the proposed system introduces a structured approach for QR-less merchant identification, several practical challenges must be considered due to limitations in real-world environments and existing infrastructure.

## 1. Location Accuracy Constraints

Accurate user positioning remains a fundamental challenge. GPS-based location systems can exhibit errors ranging from a few meters to several tens of meters, particularly in dense urban environments or indoor settings. Since the system relies on fine-grained spatial mapping(e.g., DIGIPIN cells of approximately 4m x 4m), such inaccuracies can lead to incorrect mapping of the merchant identification.

## 2. Spatial Ambiguity within DIGIPIN Cells

Although DIGIPIN reduces the search area to a smaller spatial region, multiple merchants may still exist within the same cell. This is especially common in high-density commercial areas such as shopping complexes, food courts, and multi-vendor spaces. As a result, spatial mapping alone is insufficient to uniquely identify a merchant.

## 3. Multi-Level(Vertical) Ambiguity

DIGIPIN-based systems primarily operate in a two-dimensional spatial framework and do not inherently distinguish between floors of a building. In multi-store environments, merchants located on different levels may share the same or overlapping spatial representation, introducing additional ambiguity in identification.

## 4. Variability in Contextual Signals

The system may rely on contextual inputs such as device motion, environmental signals, or user interaction patterns. However, these signals can be inconsistent or noisy depending on device conditions, user behavior, and environmental factors. This variability can impact the reliability of context-based merchant identification.

## 5. Limitations of UPI Deep Linking

While UPI deep linking enables redirection to payment applications, it does not guarantee successful transaction completion in all cases. Certain merchants rely on dynamic or backend-generated QR codes, and reconstructing equivalent payment parameters may not always be feasible. This limits the system’s ability to fully replicate QR-based transactions in all scenarios.

# Proposed Approach

To address the challenges identified in the previous section, the proposed system adopts a multi-stage approach that combines spatial mapping with contextual evaluation to improve merchant identification accuracy.

## 1. Spatial Reduction using DIGIPIN

The system utilizes DIGIPIN-based mapping to reduce the search space from a broad geographic region to a fine-grained spatial cell(approximately 4m x 4m). This step serves as the primary filtering mechanism by limiting the set of candidate merchants to those associated with the identified cell.

## 2. Contextual Refinement Layer

Since spatial mapping alone is insufficient to uniquely identify a merchant, the system introduces a contextual refinement stage. In this stage, additional signals derived from the user's device and interaction context are considered to differentiate between multiple candidate merchants within the same region.

These contextual inputs may include relative movement patterns, temporal context, and interaction behavior, which collectively help in narrowing down the most relevant merchant.

## 3. Merchant Ranking Mechanism

The system evaluates candidate merchants based on a set of relevance criteria and assigns a relative likelihood score to each merchant. This ranking mechanism enables the system to predict the most probable merchant associated with the user's intent while retaining flexibility to handle ambiguity.

## 4. Assisted Selection and Fallback

To account for uncertainty, the system may present a small set of top-ranked merchants to the user for confirmation instead of enforcing a single automatic selection. This ensures reliability while maintaining user control in ambiguous scenarios.

Overall, this approach combines spatial reduction and contextual reasoning to address the limitations of purely location-based systems while maintaining practical compatibility with existing digital payment ecosystems.

# Conclusion and Future Scope

This work presents a structured approach for enabling QR-less payments through intelligent merchant identification. By combining spatial mapping techniques such as DIGIPIN with contextual evaluation, the proposed system aims to reduce dependency on manual QR code scanning and improve the overall payment experience in dense and dynamic environments.

The system introduces a two-stage identification process, where spatial reduction limits the search space and contextual signals assist in resolving ambiguity among multiple merchants. This layered approach highlights the potential of integrating location intelligence with decision-making mechanisms in real-world digital payment systems.

In addition to improving payment interaction, the system demonstrates how inferred merchant context can be utilized to enable automatic expense classification and basic spending insights. This extension illustrates the broader applicability of contextual intelligence beyond transaction initiation.

While the current approach focuses on high-level system design and feasibility, several opportunities exist for further enhancement. Future work may explore advanced indoor positioning techniques, improving handling of vertical ambiguity, and more robust contextual modelling for merchant prediction. Integration with additional data sources and infrastructure can further improve accuracy and reliability.

From a broader perspective, the proposed system demonstrates the potential for building intelligent interaction layers on top of existing digital public infrastructure. With further development and validation, such approaches could contribute to more seamless, efficient, and context-aware payment experiences.